TEXORA · LECTURE SHEET

Spacer Fabric

Two fabric faces held apart by a resilient forest of pile yarns: a textile sandwich panel made in one process, whose thickness, compression behaviour and air and moisture transport are engineered through the spacer yarns.

FM-4103 · SPECIAL FABRIC PRODUCTION · TOPIC 2 · FABRIC MANUFACTURING SPECIALIZATION

1. Definition and structure

A **spacer fabric** is a three-dimensional textile consisting of **two independent face fabrics** (top and bottom) connected and held apart by a third yarn system — the **spacer (pile) yarns** — which run back and forth between the faces, usually as stiff monofilaments standing nearly perpendicular to them. The result is a genuine sandwich: two skins and a fibrous core, formed in a single machine pass, with no adhesive and no lamination step. Thickness is a machine setting: weft-knitted spacers are typically a few millimetres thick, warp-knitted (double-needle-bar Raschel) spacers are commonly made from about 2 mm up to several tens of millimetres, and woven spacers occupy the thinner end.



Figure 1 — Spacer fabric in cross-section (schematic): two faces joined by spacer yarns that zig-zag or step between them. Under compression the spacer yarns buckle elastically and spring back — the source of the fabric's cushioning and of its recovery advantage over foam.

2. Production routes

- **Warp-knitted spacer (the main route):** double-needle-bar Raschel machine. Two ground guide bars knit the two faces on the two needle bars independently; one or two pile bars lay spacer yarns alternately into the front and back faces, so each spacer yarn is anchored in both. Distance between

the needle bars sets thickness. Face structures are chosen per duty — closed faces for support, mesh/open faces where air flow is the point.

- **Weft-knitted spacer:** double-jersey (rib/interlock) circular or flat machines: faces knitted on cylinder and dial, connected by tuck-stitched spacer yarns (usually monofilament) fed between the beds. Thinner and softer than Raschel spacers, used in garments and footwear.
- **Woven spacer:** two woven grounds woven simultaneously on a double (velvet-type) loom, joined by pile warp, and kept apart rather than slit — the velvet machine stopped before the knife. Mostly thinner, firmer products and preforms.

Raw materials: polyester dominates (monofilament spacer yarns for resilience; textured or spun yarns in the faces), with polyamide, polypropylene and glass or metal-containing variants for technical uses. The spacer yarn's stiffness is the design core: a monofilament of a few tenths of a millimetre behaves as a tiny column — it resists until it buckles, then recovers because the polymer is elastic at these deflections.

3. Properties — and why the structure gives them

- **Compression and recovery:** load is carried by thousands of spacer columns per square metre; they buckle progressively, giving a long, soft initial compression then a firming response, and they recover on unloading — unlike polyurethane foam, the structure does not take a permanent set or crumble with age, and it survives washing.
- **Air permeability and ventilation:** the core is mostly air and the faces can be open meshes: air moves through and along the panel, carrying heat away — the reason spacers replaced foam in car seats and mattress toppers, where the foam's closed cells trap heat.
- **Moisture management:** vapour and even liquid pass along the core; spacer yarns of hydrophobic polymer do not absorb, so the panel dries fast.
- **Pressure distribution:** the panel spreads point loads over an area — the clinical property exploited in pressure-relief supports and orthopaedic braces.
- **Recyclability:** an all-polyester spacer is a single-polymer product, recyclable in principle where a foam–fabric laminate is not.

> **WORKED EXAMPLE — MASS OF A SPACER PANEL FROM ITS PARTS**
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> A warp-knitted spacer: top face 150 g/m², bottom face 150 g/m², spacer-yarn contribution 90 g/m². Total areal mass = 150 + 150 + 90 = **390 g/m²** at a thickness of, typically, 6–10 mm for such a construction.
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> Compare the function, not just the mass: a foam layer of equal cushioning thickness would weigh

> less on paper but adds a lamination step, an adhesive, and a heat-trapping closed-cell core; the spacer's 390 g/m² buys the air channel and wash durability as part of the same number.

Limitations, stated honestly: spacers cost more per square metre than the foams they replace; thick panels are bulky to transport; the cut edge shows the open core and may need binding; and the compression curve, while resilient, is softer than a firm foam at equal thickness — the design must be chosen per load case, not assumed superior.

4. End-uses

Automotive seat inserts and headliners; mattress toppers and pressure-relief overlays; medical and orthopaedic supports, braces and shoe insoles; footwear uppers (the engineered mesh shoe is largely a spacer product); sports padding and helmet liners; backpack back panels; bra cups and shoulder pads; acoustic and filtration layers; and, in glass or aramid versions, lightweight composite cores.

Glossary

Spacer yarn — the yarn system connecting the two faces and setting thickness and compression behaviour. **Double-needle-bar Raschel** — warp-knitting machine with two needle bars knitting two faces simultaneously. **Monofilament** — a single continuous filament used as a yarn. **Compression set** — permanent thickness loss after loading; low in spacers, the classic failure of foams.

Key points

• Two faces + spacer yarns, one process, no adhesive: a textile sandwich panel.
• Properties are structural: buckle-and-recover spacer columns give cushioning; the open core gives ventilation and moisture transport that foam cannot.
• Thickness and compression are machine settings (needle-bar distance, spacer yarn stiffness and density) — designed, not incidental.
• Worked panel: 150 + 150 + 90 = 390 g/m²; the honest comparison with foam includes durability, washing and heat, not mass alone.

Practice

1. Why is a monofilament, rather than a spun yarn, the usual choice for the spacer system? Answer through the column-buckling mechanism.
2. A seat insert must be washable and must not trap heat. Argue the spacer-versus-foam choice in three structure-based points.

3. State two machine variables on a double-needle-bar Raschel that set (a) thickness and (b) compression hardness, independently.
4. Compute the total mass of a spacer whose faces are 180 and 140 g/m² and whose spacer yarns contribute 75 g/m².

Answer key

1. The spacer yarn works as a slender column in compression. A monofilament is a solid rod: it has bending stiffness along its whole length, buckles elastically at a repeatable load and springs back. A spun yarn is a bundle of fibres with air between them — under axial load it collapses and compacts rather than buckling elastically, so the panel would thin and take a set.
2. (i) **Washability:** the spacer is one knitted structure with no adhesive bond to fail and no foam to crumble; it can be laundered whole. (ii) **Heat:** the core is an open air channel and the faces can be meshes, so body heat is convected away; foam's closed cells insulate and trap it. (iii) **Durability of cushioning:** spacer columns recover elastically cycle after cycle, where foam progressively takes compression set — the seat stays supportive after washing and ageing.
3. (a) Thickness: the **distance between the two needle bars** (trick-plate/bed separation). (b) Hardness: the **spacer yarn itself — its diameter/material stiffness and its density of insertion (how many spacer yarns per unit area), together with its inclination** (more vertical = harder).
4. 180 + 140 + 75 = **395 g/m²**.
